



VRender™ 3D Graphics Application Processor For 3D Handheld Appliances

VRender™ 3D

Overview

VRender™ 3D(VR3511F) is the 2nd generation chip adopting MagicEyes' VRender Technology. Its high performance and low power consumption architecture realize a big differentiation in developing a high-end 3D graphics embedded applications such as 3D Car Navigation System, 3D pachinko & arcade game system, 3D MP3 player, Portable 3D Game Player and 3D Mobile Commerce. VRender 3D is a real SoC including ARM920T 32bit CPU, 3D Graphics Accelerator, Video Processor, 64Poly Wavetable Synthesizer and various interfaces like USB, UART and AC97.

The embedded ARM920T CPU, which contains 16KB I-Cache, 16KB D-Cache, and MMU is suitable for implementing various kinds of OS like Windows CE, μ C OS, Linux and a broad application software as well.

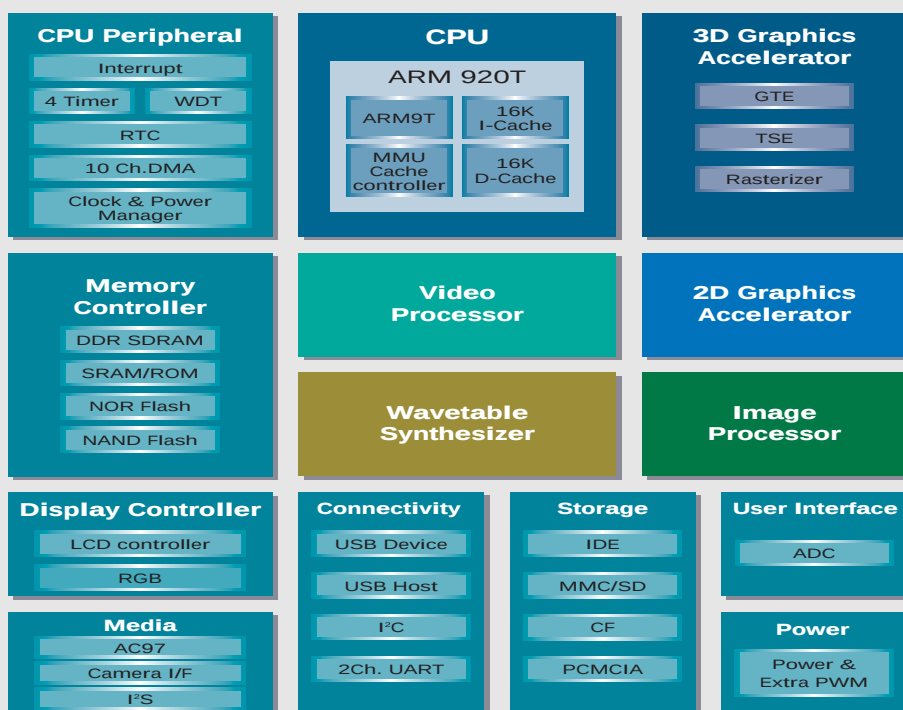
The embedded 3D Graphics engines, such as Triangle Setup Engine, Rendering Engine and Geometric Transform Engine, reduce CPU overhead dramatically and provide high performance as well as Tremendous graphics functionality under low power consumption.

Geometric Transform Engine is consisted with 4 floating engines. In existing 3D applications, CPU consumes most of CPU performance to calculate Geometric arithmetic. VRender 3D's Geometric Transform Engine performs Geometric arithmetic instead of CPU to reduce CPU overhead. It enables high performance 3D graphics in restricted CPU performance environment such as Mobile application.

VRender 3D supports various memory cards - SD/MMC, CF. It also supports interfaces like UART, I²C, USB, IDE and AC97. This kind of wide interface enables VRender 3D to use minimal interface chip and glue logic.

VRender 3D supports DDR SDRAM. Supporting DDR SDRAM makes VRender 3D to have a more powerful 3D graphics performance.

VRender™ 3D Block Diagram



Key Features

- Embedded ARM920T
- 3D Graphics GTE (Geometric Transform Engine)
- 3D Graphics Rasterizer
 - 1,000,000 Polygon/sec
- Wavetable Synthesizer
 - 64 poly
- Video Processor
 - MPEG4 Decode 720×480, 30fps

VRender™ 3D

Features

Technology

- 0.18um CMOS, 1.8V Core, 3.3/2.5V I/O
- PKG : 400pin FBGA (18mm x 18mm)

ARM920T CPU

- 16KB Instruction Cache
- 16KB Data Cache
- Operation Frequency : 200MHz

3D Graphics GTE (Geometric Transform Engine)

- 4 Floating Vector Engines.
 - Based on 24bit Floating Point
 - Floating Point Adder
 - Floating Point Multiplier
 - Floating Point Divider
 - Floating Point Reciprocal
- 64 Depth Program Memory
- 16 Vector Input Registers
- 16 Vector Constant Value Registers
- 32 Vector General Purpose Registers
- World / Camera Transform
- Bezier Curve Algorithm, Vertex Interpolation
- Lighting
 - Ambient Light
 - Point Light
 - Directional Light
- Clipping by 6 Planes

3D Graphics Rasterizer

- Sub-Texture Accuracy
- Sub-Pixel Accuracy
 - 2x2 Sub-pixel Accuracy
- Perspective Corrected Multi Texture Mapping
 - Color Format : 4 / 8 / 16 Formats
 - Adaptive Bi-linear Filtering
 - Use MPEG Video Texture
 - MIPMAP
 - Multi-Texture
 - Support Tiled Texture mapping
 - Internal Texture Cache :
8Kbyte(64x64pixel@8bpp)x2 way
- Specular / Diffuse Lighting
- Fogging
- Support Enhanced Z-Buffer

- Alpha Blending compatible with Direct3D
- Hardware Dithering
- FSAA(Full Screen Anti-Aliasing) : 1 x 2 Mode

Video Processor

- Hardware Accelerators
 - IDCT Hardware Accelerator
 - IQ Hardware Accelerator
 - Motion Prediction & Compensation Hardware Accelerator
 - CSC(Color Space Converter)
 - Huffman Decoder & Encoder
 - Bit Parsing Processor
- Performance
 - MPEG4 SP Decode 720x480, 30fps

Wavetable Synthesizer

- Based on Wavetable Method.
- Capable of producing up to 64 different sounds simultaneously
- Programmable 6 Output Channels.
- Individual Panning, Volume, Envelop, Tremolo, Vibrato
- Sample Formats
 - 8bit PCM, 16bit PCM, 8bit u-LAW

Image Processor

- Direct Interface with CMOS sensor

Display

- 16bit color, Color Space Converting, Scaling filter, etc.,
- 4 Layer Overlay
- Resolution : 1024 x 768

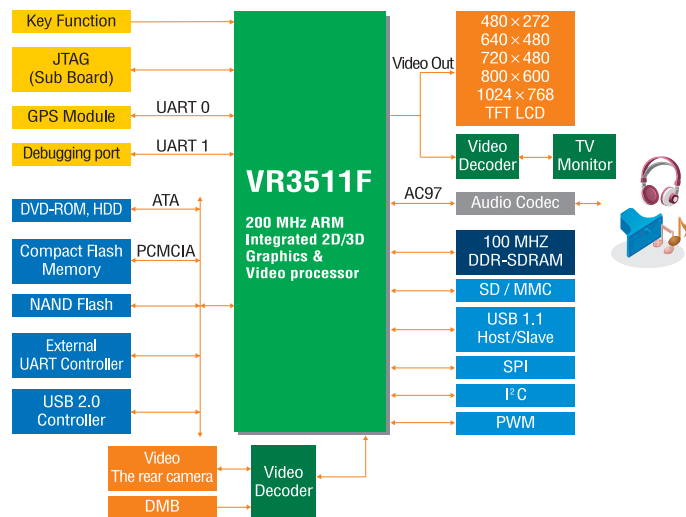
Memory

- DDR SDRAM
- SRAM / ROM
- NAND / NOR Flash
- PCMCIA / CF

Various Interfaces

- USB host / device, IDE, I²C, I²S, UART, AC97, etc

VRender™ 3D Car Navigation System Diagram



MagicEyes...

Since MagicEyes, headquarter in Korea, was established in 1997 as a high-end SoC & multimedia system solution provider, it has introduced several chips targeted for digital-convergence and digital ubiquitous systems. Now MagicEyes is recognized as one of leading companies in multimedia SoC market though its outstanding technical achievements.